CS 225 Examination

Data Structures and Algorithms in C++

Instructions:

- Answer all questions.
- For Questions 1–4, choose the best option.
- For Questions 5–8, write brief but precise answers.

1. What is the time complexity of searching for an element in a balanced binary search tree with n nodes?

(A) $O(\log n)$

(B) $O(n)$

(C) $O(n \log n)$

(D) $O(1)$

2. In C++, which of the following correctly describes the difference between a shallow copy and a deep copy?

(A) A shallow copy copies pointer values; a deep copy allocates new memory and copies the data

(B) A shallow copy is faster; a deep copy is always preferred

(C) Both create independent copies of all data members

(D) A deep copy only copies primitive types

3. Which data structure would be most appropriate for implementing a function call stack in a programming language runtime?

(A) Queue

(B) Circular buffer

(C) Stack

(D) Hash table

4. What is the worst-case time complexity of QuickSort when the pivot selection is poorly chosen?

(A) $O(n \log n)$

(B) $O(n^2)$

(C) $O(n)$

(D) $O(\log n)$

5. Explain the concept of amortized analysis. Use the dynamic array (vector) resizing operation as an example to show why insertion has $O(1)$ amortized time complexity despite occasional $O(n)$ resize operations.

6. Describe the differences between a min-heap and a priority queue. How would you implement Dijkstra's shortest path algorithm efficiently using a priority queue, and what would be the time complexity?

7. Explain the principles of the Rule of Three in C++ and why it's important for classes that manage dynamic memory. What are the Rule of Five and Rule of Zero, and when should each be applied?

8. Compare the advantages and disadvantages of using std::vector versus std::list in C++. Under what circumstances would you choose one over the other, considering factors like memory layout, cache performance, and insertion/deletion patterns?